Bank of America Project Lab
AI HQLA Risk-Optimization

December 17th, 2025

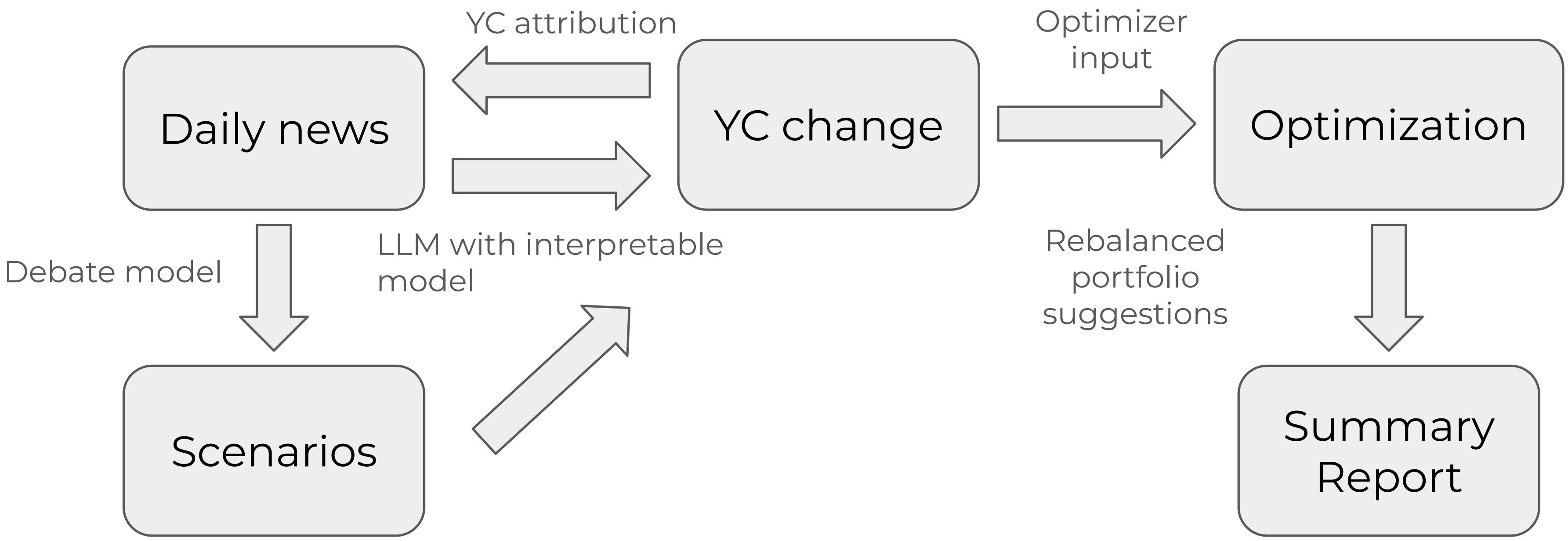
BANK OF AMERICA

Risk Pipeline

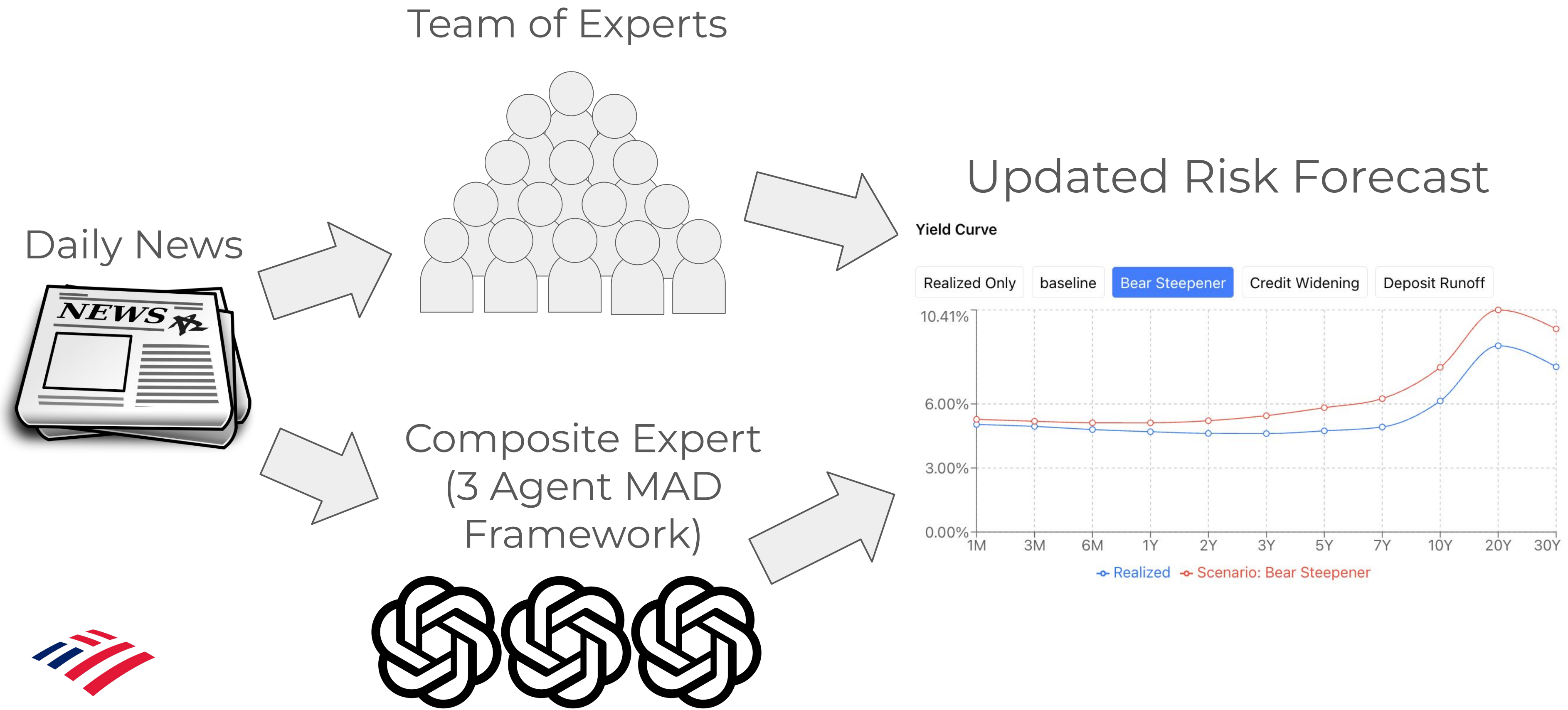
What do we think is going to happen? → Should any scenario occur, what will the state of the world be and how should we position ourselves? → How do realized events update our predictions?

Our dashboard proposal identifies where LLM capabilities can bolster each step of this pipeline

Approach Outline



Identifying Plausible Risk Scenarios



Data Source and News Ingestion

- Yahoo
- FT
- CNBC
- Reuters
- …

Web crawler →

8 buckets

- Monetary_policy
- Economic_data
- Geopolitical_events
- Market_sentiment
- Fiscal_policy
- Credit_events
- Commodity_prices
- other_general

LLM →

Useable data set

Multi agent Debate

Prompt

"You are given a portfolio representing Bank of America's HQLA holdings. Construct a set of plausible macro-financial risk scenarios over a six-month horizon and assign probabilities to each scenario."

Portfolio Input

- Asset mix: USTs, IG Credit, HY Credit
- Duration: 4.8 years
- Funding: 65% retail deposits
- Constraints: LCR ≥ 110%, CET1 ≥ 11%

Debate

Based on the portfolio and current macro economic trends I suggest the following scenarios …

I disagree about your first scenario because …

I understand your concerns and I suggest …

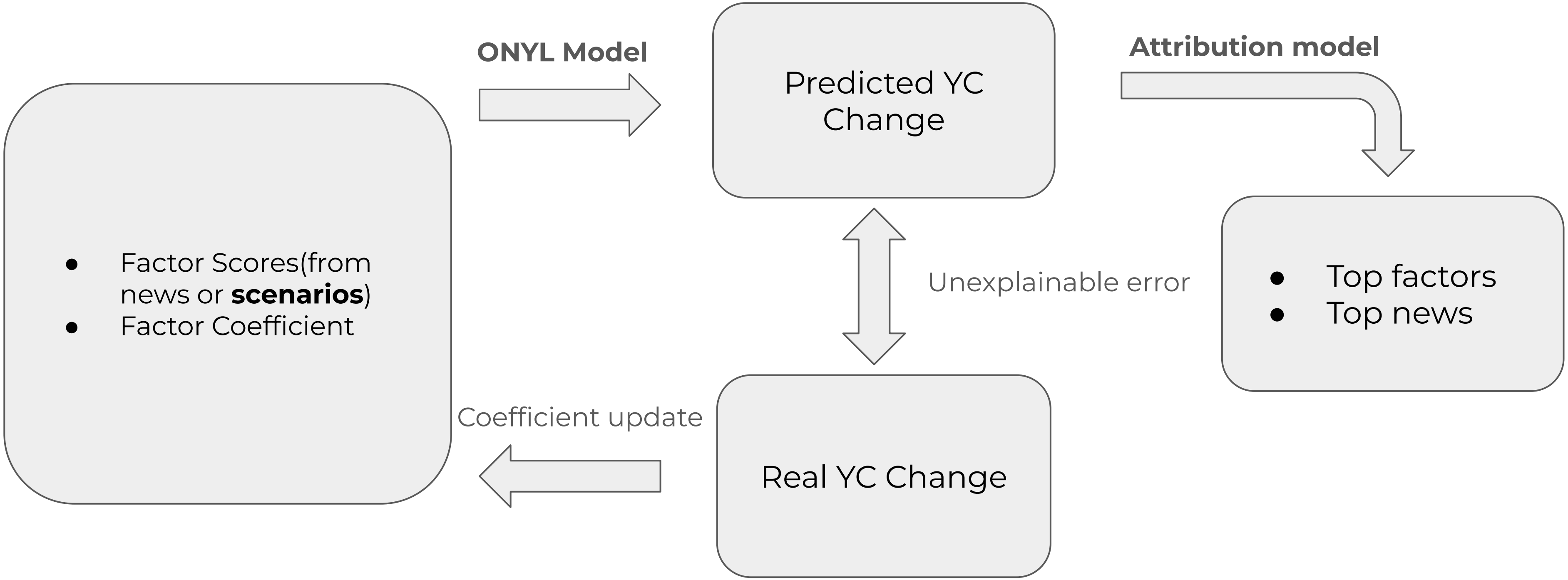
…

After considering both sides, I believe that the final scenarios should be …

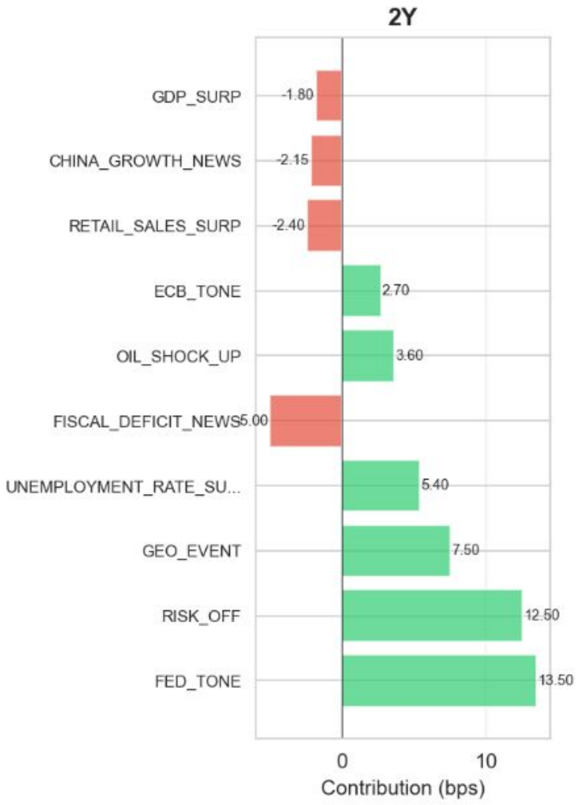
Scenario Generation Matrix

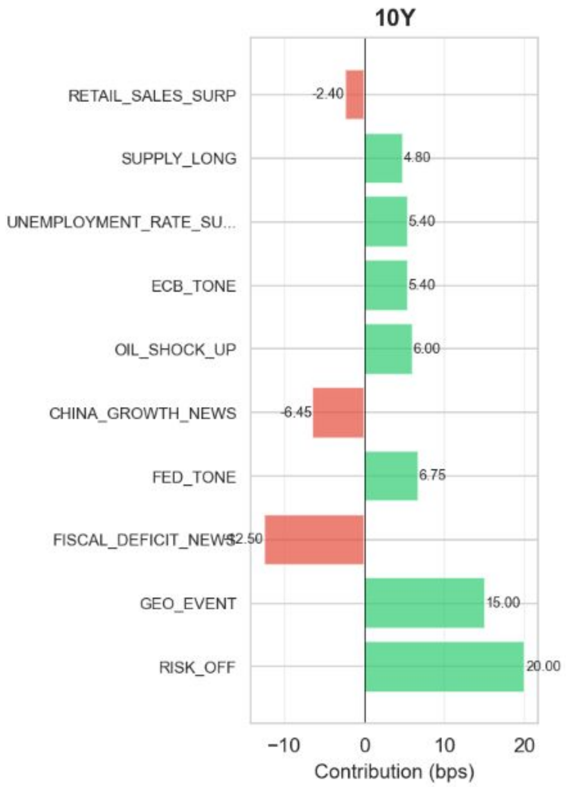
Scenario	**Macro Driver**	**Probability**	**Key Shock**	**Risk Channels**	**Signals**
Bear Steepener	Inflation persistence	20%	Long Rates +50bp	Rates	10Y > 4.0%, CPI > 3.5%
Credit Widening	Risk Aversion	15%	IG +15bp, HY +45bp	Credit	IG OAS > 135bp
...	...	...	...	...	...

Yield Curve Prediction and Attribution



Yield Curve Prediction and Attribution





Yield Curve Prediction and Attribution

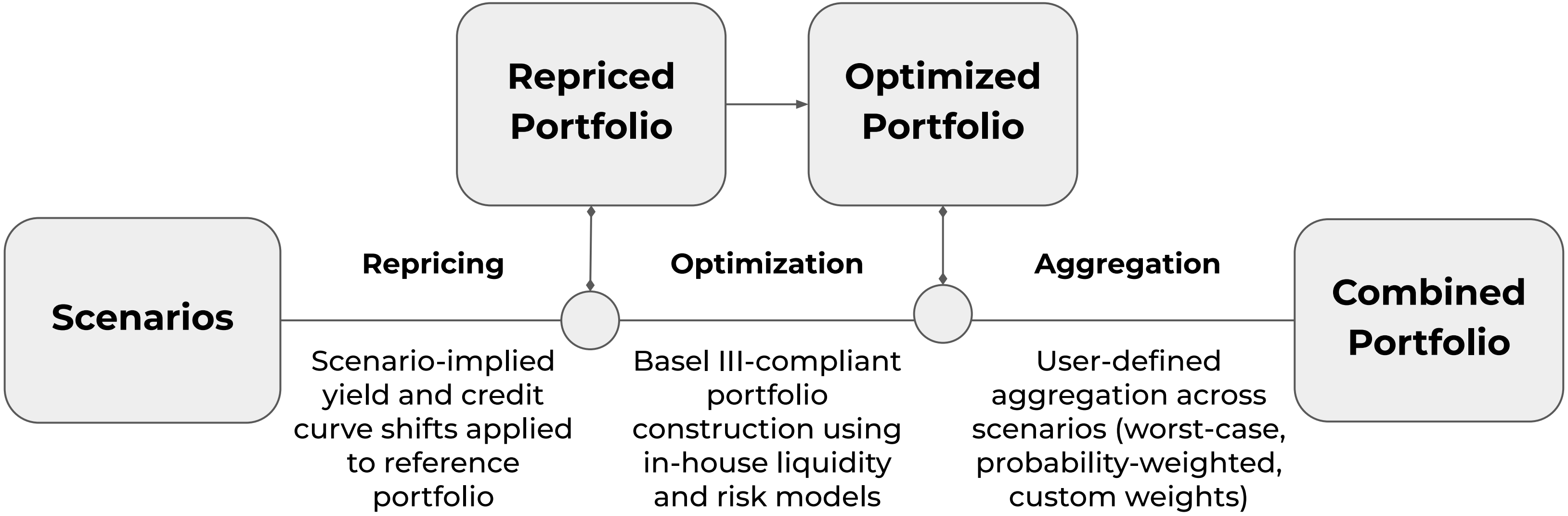
News Attribution

Top news-driven factors and their key headlines for the selected date.

1. RISK_OFF	-2.500
China Vanke's bonds plunge on fears of waning state support	-1.35
China industrial profits drop 5.5% in October, worst performance in five months	-0.80
Gold prices cool after surging on rate cut cheer, Fed Chair speculation	-0.70

Similar columns for each significant news factor.

Scenario Based Optimization



AI-Generated Portfolio Summary

Purpose: Improves transparency and frames key directional considerations

LLM Inputs

- Scenarios
- Probabilities
- Optimized Portfolios

AI Interpretation and Synthesis

Summary Report

- Executive summary
- Stress behavior
- Final Portfolio

Daily Deliverables

- **Scenario matrix**
  - Structured mapping of macroeconomic and market scenarios to assigned probabilities
- **Key news drivers**
  - Most impactful news events driving yield-curve movements within each news bucket
- **Portfolio summary**
  - Snapshot of portfolio positions, liquidity metrics, and scenario-based allocations

Thank you for listening!

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Appendix

Sample Output: MAD Scenario Generation Matrix

Scenario Generation

Scenario	Description	Probability	Rationale	ImpactChannels	Shocks	MetricsDelta	TradeList	Assumptions	PredictionDate	probability_sum_before	Signals
Bear Steepener	Yield curve steepens with long-term rates rising 50 bps while short-term rates remain stable.	20%	Long-term inflation expectations are rising due to sustained consumer spending, leading to increased Treasury yields.	Rates	{"move_index":50,"yield_curve":"bear_steepener"}	{"LCR":-2,"NII":-5}	Add $1bn 10-year Treasuries, Reduce $500mn short-term repos	Continued consumer spending supports growth; Fed remains cautious on rate hikes.	2025-12-16	1	10-year Treasury yield > 4.00%, CPI > 3.5% YoY, FOMC meeting on 2024-01-31
Credit Widening	Widening of IG and HY spreads by 15 bps and 45 bps respectively, impacting corporate bond valuations.	15%	Increased market volatility leads to a flight to quality, raising risk premiums across credit markets.	Credit	{"credit_spreads":{"ig_oas":15,"hy_oas":45}}	{"LCR":-3,"OCI":-10}	Sell $500mn of underperforming corporate bonds, Increase cash reserves by $1bn	Economic uncertainty leads to higher risk aversion among investors.	2025-12-16	1	IG OAS > 135 bps, HY OAS > 455 bps, Corporate earnings reports in early February
Deposit Runoff	Increase in retail deposit beta leading to a 5% runoff in retail deposits, tightening liquidity.	20%	As interest rates rise, depositors seek higher returns, potentially shifting funds to higher-yielding alternatives.	Deposits	{"deposits":-5}	{"LCR":-5,"NSFR":-3}	Increase wholesale funding by $2bn, Issue $1bn of longer-term CDs	Deposit competition heats up as rates rise.	2025-12-16	1	Retail deposit beta > 30%, SME deposits decline > $3bn by March 2024
Regulatory Tightening	Increased TLAC requirements and changes in liquidity add-ons affecting capital ratios.	15%	Regulatory bodies push for higher capital buffers amid economic uncertainty, leading to tighter liquidity positions.	Regulation	{"regulatory_changes":"Increased TLAC requirements"}	{"LCR":-2,"CET1":-10}	Reallocate $1bn to Tier 1 capital instruments, Hold additional cash reserves of $500mn	Regulatory environment shifts towards more conservative capital requirements.	2025-12-16	1	Final Basel Endgame rules released by 2024-02-15, Public comments on TLAC proposals by 2024-03-01
Stable Economic Growth	Stable economic growth with GDP growth at 2.5% and no major shocks to the system.	30%	Continued consumer spending and a stable job market support growth, keeping rates in check.	Rates	{"move_index":0}	{"LCR":3,"NII":5}	Add $1bn of 5-year Treasuries, Reinvest $500mn of maturing assets	Inflation remains under control; Fed maintains a balanced approach to interest rates.	2025-12-16	1	GDP growth > 2.5% in Q1 2024, Unemployment rate stable < 4%

Appendix

ONYL formulas:

1. Overview

ONYL is a daily-updating linear model that links **news-derived macro factors** to **yield curve changes**, using online gradient-style coefficient updates.

Each factor has a defined macro meaning (e.g., *FED_TONE, SUPPLY_LONG, INFLATION_NEWS*), and each tenor learns a separate sensitivity to every factor.

Goal:

> Predict daily yield curve changes and attribute movements to economic factors.

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2. Prediction Equation

For each tenor (k) and day (t):

$$\Delta\hat{y}_{t,k} = b_k + \sum_f B_{k,f}\, x_{t,f}$$

Where: [

$$\begin{aligned} x_{t,f} &: \text{ factor score for factor } f \text{ on day } t \\ B_{k,f} &: \text{ coefficient (bps per unit factor)} \\ b_k &: \text{ intercept} \\ \Delta\hat{y}_{t,k} &: \text{ predicted change in yield (bps)} \end{aligned}$$

]

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3. Error

$$e_{t,k} = \Delta y^{\text{obs}}_{t,k} - \Delta\hat{y}_{t,k}$$

4. Update Weights

Each factor gets an adaptive weight (w_{t,f}):

$$w_{t,f} = \begin{cases} 0.3, & |x_{t,f}| < 0.3 \\ \text{scaled so that } \sum_f |w_{t,f} x_{t,f}| \le 3.0, & \text{otherwise} \end{cases}$$

This prevents single large-score factors from dominating the update.

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5. Coefficient Update Rule

Unclipped update:

$$\Delta B_{k,f} = \eta\, w_{t,f}\, e_{t,k}\, x_{t,f}$$

With update clipping:

$$|\Delta B_{k,f}| \le 0.8 \text{ bps}$$

Final update:

$$B_{k,f} \leftarrow B_{k,f} + \Delta B_{k,f}$$

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6. Sign Guards (Economic Constraints)

Example constraints:

- *SUPPLY_LONG* → raises long-end yields → coefficients must be **non-negative**
- *RISK_OFF* → lowers yields → coefficients must be **non-positive**

Enforced by projection:

$$B_{k,f} \leftarrow \max(B_{k,f},\, 0) \quad \text{(for non-negative guard)}$$

Optimization Framework

Optimization Architecture

- **Level 1: Basel-Constrained Feasibility**
  - Enforces LCR ≥ minimum, haircut limits, level caps, and allocation bounds; Ensures regulatory compliance

- **Level 2: Lexicographic Mean–Variance Optimization**
  - Follows Moody's mean-variance framework
  - Objective 1: maximize expected portfolio return, proxied by YTM
  - Objective 2: control variance subject to Level 1 feasibility

Scenario Workflow

- Start from the current portfolio and a reference portfolio
- Reprice each portfolio under the yield-curve scenarios provided by yield curve predictions
- Optimize allocations per scenario using:
  - Mean-lexicographic
  - Mean–variance-lexicographic
- Combine scenario-optimal portfolios using:
  - Probability-weighted aggregation
  - Worst-case selection
  - Top-K worst average (by expected return or LCR)

Note: Scenario generation, pricing, and optimization logic are fully modular so the optimization can be replaced with in-house models